MIRAGE: Optimal, Road-Graph-Native Location Obfuscation at City Scale for IoT Smart Cities

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Abstract—IoT smart cities depend on continuous location reporting, and a large family of obfuscation mechanisms—k-anonymity, differential privacy (DP), graph constraints, temporal cloaking—has been proposed to protect it. Two questions remain unanswered in a deployable, quantitative way. First, how much privacy does each mechanism actually provide, on a common, comparable scale and against the threats it faces? Second, how far are these heuristics from the best achievable privacy on a real road network? We answer both. (1) We adopt an adversary-grounded privacy metric—the expected error of an optimal Bayesian adversary, in metres—and instantiate it under two threat models, a snapshot adversary and a Viterbi trajectory adversary with a Markov mobility prior, showing that a single privacy scale is inadequate because the two adversaries rank mechanisms differently. (2) We introduce MIRAGE, the first road-graph-native optimal obfuscation mechanism that scales to a real city. MIRAGE computes, per location, the release distribution that provably maximises the Bayesian adversary's error subject to a utility budget, with the distortion and adversary error measured as graph shortest-path distance; it is made tractable by solving one small linear program per density-adaptive local region, turning an intractable $|V|^2$ program into $|V|$ tiny ones. On real OpenStreetMap networks of two cities under three mobility datasets (Beijing: GeoLife, T-Drive; Porto: taxi) we measure the price of heuristics: at matched utility, DP and k-anonymity leave 5–22% of the achievable privacy on the table in the operationally relevant regime, while MIRAGE is Pareto-optimal (privacy$\approx$utility) up to the network's intrinsic uncertainty ceiling. We explain the cross-city variation through the prior heterogeneity that the optimal mechanism exploits and heuristics ignore. Finally, we show—honestly—that snapshot-optimality does not imply trajectory-optimality: MIRAGE's scalability partition leaks region membership, so against a sequential adversary a diffuse mechanism can be more robust, which identifies the key open problem. The complete framework is released for reproducibility.

Index Terms—Location privacy, optimal obfuscation, Bayesian adversary, differential privacy, k-anonymity, road networks, trajectory privacy, linear programming, Internet of Things, smart cities.

1 Introduction

Smart cities instrument roads, vehicles, and people with IoT devices that report location continuously for analytics, traffic management, and emergency response. The same reports expose homes, workplaces, health-facility visits, and detailed mobility, even after identifiers are stripped [1], [2]. Two decades of research have produced diverse obfuscation mechanisms: spatial k-anonymity [1], [2], [10], differential privacy (DP) and geo-indistinguishability [3], [8], graph-constrained perturbation [7], density-aware adaptation [12], and temporal cloaking [2].

Despite this richness, a practitioner deciding what to deploy on a real road network faces two questions that the literature does not answer in a deployable, quantitative way.

Q1: How private is each mechanism, comparably and against the right threat? Mechanisms are scored on incompatible, family-specific scales—k, a log-scaled privacy budget ε, cloaking-region area, information loss—that carry no common operational meaning. A cloaking region of 500 m radius and a Laplace noise of 500 m standard deviation are not the same privacy. Worse, mechanisms are almost always evaluated against a single snapshot attacker, even those (like temporal cloaking) whose purpose is to defeat trajectory tracking. Without a common, threat-aware scale, "which is more private?" has no defensible answer.

Q2: How far are deployed heuristics from the best achievable privacy? Classical theory shows that, for a given utility budget, there exists an optimal obfuscation mechanism—the solution of a linear program (LP)— that maximises a Bayesian adversary's expected error [5]–[7]. But that theory has only ever been instantiated on the Euclidean plane or small grids, is snapshot-focused, and does not scale (the LP has $|V|^2$ decision variables). It has therefore never been evaluated on a real road network, and the gap between the heuristics people actually deploy and the optimum on real topology is simply unknown.

A motivating example. A city runs 10,000 connected vehicles reporting every 10 s. The operator can spend, say, 500 m of average spatial distortion for analytics to remain useful. Which release policy buys the most privacy for that budget on this city's roads and mobility? DP with a tuned ε? k-anonymity with a tuned k? Or something provably better? Today the operator cannot compute the answer, nor know how much privacy a suboptimal choice forfeits.

We answer both questions and quantify the example. Contributions.

1) An adversary-grounded, threat-model-aware privacy metric (Section 5). Privacy is the expected error of an optimal Bayesian adversary [4], in metres, evaluated under both a snapshot adversary and a

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Viterbi trajectory adversary equipped with a Markov mobility prior. We show the two adversaries induce different rankings, so a single privacy scale is invalid.

2) MIRAGE, the first road-graph-native optimal obfuscation mechanism that scales to a real city (Section 6). MIRAGE solves the optimal-mechanism LP with graph shortest-path distortion and on-graph output support, made tractable by density-adaptive local LPs with a correctness argument and an $O(|V|C^3)$ precomputation. DP and k-anonymity arise as constrained/degenerate special cases; an optional geo-indistinguishability constraint recovers a formal ε-DP guarantee.
3) The price of heuristics on real road networks (Sections 8–9). Across three datasets and two cities we quantify, at matched utility, how much privacy DP and k-anonymity sacrifice (5–22% in the practical regime), with 95% bootstrap confidence intervals, and we explain the cross-city variation through the prior heterogeneity the optimal mechanism exploits.
4) An honest trajectory analysis (Section 13): we build a trajectory-aware variant (MIRAGE-T) and show that snapshot-optimality does not imply trajectory-optimality—the scalability partition leaks region membership— identifying the key open problem.
5) A reproducible, real-network, multi-city framework and open release spanning a grid abstraction and real OSM graphs of Beijing and Porto, three datasets, ablations, scalability measurements, and a radio-technology sensitivity analysis of the energy model.[1]

Honest positioning: we do not invent optimal obfuscation—that is due to Shokri et al. [5], [6] and Bordenabe et al. [7]. Our contribution is to make it graph-native, city-scale, and measured on real road networks, to explain when it helps, and to place it in a unified two-threat evaluation.

2 Preliminaries and Notation

We model the environment as an undirected weighted graph $G=(V,E)$ whose vertices are discrete locations (road-network nodes) and whose edge weights are road lengths; $d_G(u,v)$ denotes shortest-path distance in metres. A user is at a true node $v \in V$; a mechanism releases an observation o drawn from a conditional distribution $f(o \mid v)$, the obfuscation mechanism. We restrict outputs to graph nodes, $o \in V$, so every release is a valid on-road location.

The adversary and mechanism share public background knowledge: a population prior $\pi(v)$, the marginal probability an arbitrary report originates at v (estimated as normalised node occupancy), and a first-order Markov mobility model $P(v_{t+1} \mid v_t)$ with transition matrix $\mathbf{T}$, capturing how users move between successive time steps. Table 1 summarises notation.

Utility. We measure utility loss as expected distortion $\mathrm{U}(f)=\sum_v \pi(v) \sum_o f(o \mid v)\, d_G(v,o)$, the mean displacement of released from true locations—the quantity analytics degrade with.

Privacy. Following Shokri et al. [4], privacy is the expected error of an optimal adversary who inverts f using π (and, for trajectories, $\mathbf{T}$). This is measured in metres and is identical in form for every mechanism, so it is directly comparable—unlike k or ε.

1. https://github.com/Pushkal-Gupta/Graph-based-Location-Privacy-IoT

TABLE 1
Notation.

Symbol	Meaning
$G=(V,E)$	road-network graph; d_G shortest-path distance (m)
$\pi(v)$	population prior (occupancy) at node v
$\mathbf{T}$	Markov mobility transition matrix, $\mathbf{T}_{vw}=P(w \mid v)$
$f(o \mid v)$	mechanism: release prob. of o given true v
$D_{\max}$	utility budget: max expected distortion (m)
$\hat{v}$	adversary's Bayes estimate of the true node
C	MIRAGE local-region size
ε	(geo-ind) privacy budget (per metre)

TABLE 2
Positioning vs. closest prior work. G=graph-native metric, Sc=city-scale, RN=evaluated on a real road network, Tr=trajectory adversary measured, MC=multi-city.

	G	Sc	RN	Tr	MC
Geo-ind (Andrés'13) [8]	no	–	no	no	no
Optimal mech. (Shokri'12) [5]	no	no	no	no	no
Optimal geo-ind (Bordenabe'14) [7]	no	no	no	no	no
DP under correlation (Xiao'15) [13]	no	–	no	yes	no
MIRAGE (this work)	yes	yes	yes	yes	yes

3 Related Work

Heuristic mechanisms. Spatial cloaking and k-anonymity generalise a location to a region containing $\geq k$ users [1], [2], [10], [11]; geo-indistinguishability adds planar Laplace noise with a metric privacy guarantee [8], [9]; graph projection snaps noisy points to the road network [7]; density-aware schemes vary k locally [12]; temporal cloaking delays reports until k users co-occur [2]. All are heuristics: they fix a rule and accept whatever privacy it yields, and are typically evaluated within one family.

Optimal mechanisms. Shokri et al. [5] showed the utility-optimal mechanism against a Bayesian localisation adversary is an LP; [6] casts it as a Stackelberg game; Bordenabe et al. [7] add geo-indistinguishability as linear constraints. All of this work is on the Euclidean plane or regular grids, is snapshot-focused, and is explicitly scalability-limited (the LP has $|V|^2$ variables—$\sim 10^7$ for our graphs); none is evaluated on a real road network. MIRAGE closes exactly this gap and additionally measures the resulting benefit at city scale.

Trajectory correlation. Xiao and Xiong [13] showed per-report DP degrades under Markov temporal correlation; subsequent work handles trajectories via DP composition, dummy trajectories, or synthesis [14]–[16]. We instead use a Viterbi trajectory adversary to measure this degradation across mechanisms on a common scale, and study whether an optimal snapshot mechanism is also trajectory-robust.

Quantifying privacy. Shokri et al. [4] define location privacy as the expected error of an optimal adversary—the metric we adopt. Table 2 positions this paper against the closest prior work along the axes that matter for deployment.

TABLE 3
Datasets after map-matching to the real graphs.

Dataset	City	Points	Graph nodes
GeoLife	Beijing	∼11.4 M	3,154
T-Drive	Beijing	∼4 M	3,154
Porto (taxi)	Porto	∼81 M	3,455

4 System and Threat Models

Graphs. We use two graphs over central Beijing: a controlled 30×30 grid (900 nodes) that removes topological confounds, and the real OpenStreetMap (OSM) drive network for the same bounding box, consolidated to 3,154 nodes and 5,111 edges. Unlike the grid's uniform degree 4, the real graph has irregular degree (mean 3.24, max 23), 16% dead-ends, and road hierarchy. For a second city we build a real Porto OSM graph (3,455 nodes, mean degree 3.04, 27% dead-ends). Intersections are consolidated within a tolerance so the graph is pipeline-friendly while preserving real topology; shortest-path distances are shared by every mechanism so none is advantaged by a different metric.

Data. Three real mobility datasets across the two cities (Table 3). GeoLife [17], [18] is 182 users of multi-modal GPS in Beijing; T-Drive is ∼10k Beijing taxis over a week (a much denser regime); the Porto ECML/PKDD taxi dataset is 1.71 M trips. Each is map-matched to the corresponding real graph by projecting GPS to UTM and snapping to the nearest node. Using datasets of very different density, and a second city, separates mechanism properties from single-dataset or single-city quirks.

Two adversaries. Both know the mechanism f and the public knowledge $(\pi, \mathbf{T})$, but not the mechanism's private randomness. A snapshot adversary observes one release; a trajectory adversary observes a sequence and decodes it. We formalise both next.

5 Adversary-Grounded Privacy Metric

5.1 Snapshot Adversary

Observing a release o, the adversary forms the Bayes posterior

$$P(v \mid o) = \frac{f(o \mid v)\,\pi(v)}{\sum_{v'} f(o \mid v')\,\pi(v')}, \tag{1}$$

and outputs the estimate minimising expected graph distance, $\hat{v}(o) = \arg\min_{h \in V} \sum_v P(v \mid o)\, d_G(h, v)$. The privacy of f is the expected error of this optimal estimator,

$$\mathrm{Priv}_{\mathrm{snap}}(f) = \mathbb{E}_{v \sim \pi, o \sim f(\cdot \mid v)}\left[d_G(v, \hat{v}(o))\right]. \tag{2}$$

The observation model $f(o \mid v)$ is mechanism-specific—an anonymity set for k-anonymity, a Laplace kernel for DP, the LP solution for MIRAGE—but the measured quantity, expected localisation error in metres, is identical for all.

5.2 Trajectory Adversary

Given a sequence $o_{1:T}$ for one user, the adversary maintains a belief over the hidden true node using the mobility model. In filtering (causal) form, the belief b_t predicts through mobility and updates through the emission:

$$b_t^-(v) \propto \sum_{v'} b_{t-1}(v')\,\mathbf{T}_{v'v}, \quad b_t(v) \propto b_t^-(v)\, f(o_t \mid v). \tag{3}$$

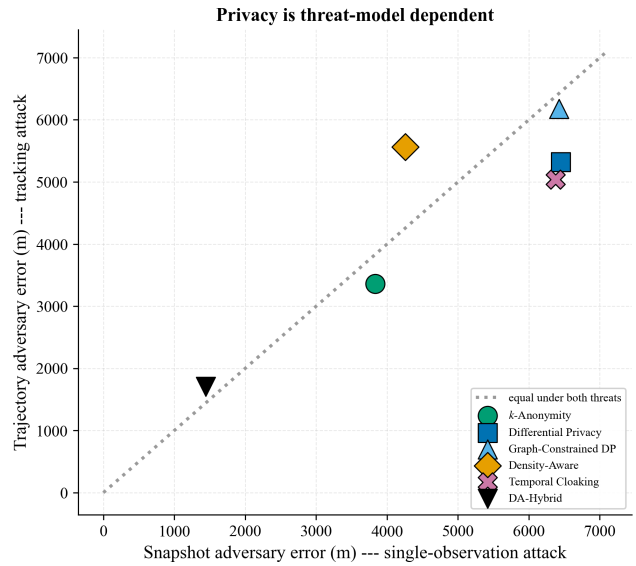

Fig. 1. Privacy is threat-model dependent: mechanisms rank differently under the snapshot (x) and trajectory (y) adversaries; points off the diagonal are protected asymmetrically.

Its per-step estimate is $\hat{v}_t = \arg\min_h \sum_v b_t(v) d_G(h, v)$; the trajectory privacy is the mean tracking error $\frac{1}{T}\sum_t d_G(v_t, \hat{v}_t)$. (A Viterbi/smoothing adversary that also uses future observations is stronger; we report the causal tracker, matched to an online attacker.) This is the threat that temporal mechanisms target and that per-report noise fails to defeat, because temporal correlation lets the adversary average noise out [13].

5.3 Why a Single Scale Is Inadequate

Fig. 1 plots the two errors against each other for the classical mechanisms on the grid. Density-aware adaptation and temporal cloaking swap rank between the snapshot and trajectory adversaries: a mechanism strong against one is weak against the other. Any single-scale privacy comparison therefore misranks mechanisms that defend different threats—motivating both the two-threat metric and, later, an honest trajectory analysis of MIRAGE itself.

6 MIRAGE: Optimal Graph-Native Obfuscation

6.1 The Optimal Mechanism on a Graph

For a prior π and a utility budget $D_{\max}$, we seek the release distribution f that maximises the snapshot adversary's expected error (2) subject to the utility budget. The adversary's expected error, given observation o, is a minimisation over its estimate h; introducing one auxiliary variable y_o per observation to encode that inner minimum linearises the problem [5], giving the LP

$$\max_{f \geq 0, y} \sum_o y_o \quad \text{(privacy, metres)} \tag{4a}$$

$$\text{s.t.} \quad y_o \leq \sum_v \pi(v) f(o \mid v)\, d_G(h, v) \quad \forall o, h \quad \text{(adversary)} \tag{4b}$$

$$\sum_v \pi(v) \sum_o f(o \mid v)\, d_G(v, o) \leq D_{\max} \quad \text{(utility)} \tag{4c}$$

$$\sum_o f(o \mid v) = 1 \quad \forall v. \quad \text{(stochastic)} \tag{4d}$$

Algorithm 1 MIRAGE precompute (per graph, offline)

Require: graph G, prior π, budget $D_{\max}$, region size C
partition V into local regions (greedy, densest-first)
for each region R **do**
    $d_R \leftarrow$ graph distances on R; $\pi_R \leftarrow \pi|_R$ (renorm.)
    $f_R \leftarrow$ solve LP (4) on $(d_R, \pi_R, D_{\max})$
    store $f_R(\cdot \mid v)$ for each $v \in R$
end for

At optimality each y_o equals $\min_h \sum_v \pi(v) f(o \mid v) d_G(h,v)$, so $\sum_o y_o$ is exactly $\mathrm{Priv}_{\mathrm{snap}}(f)$; maximising it yields the privacy-optimal mechanism for the budget. Distances d_G are graph shortest-paths and outputs are graph nodes, so the mechanism is native to the road network—new relative to the Euclidean/grid formulations of [5], [7]. An optional geo-indistinguishability constraint

$$f(o \mid v) \le e^{\varepsilon d_G(v,v')} f(o \mid v') \quad \forall o, v, v' \tag{5}$$

adds a formal ε-DP guarantee, recovering [7] as a constrained special case; k-anonymity (uniform release over the k nearest) and plain geo-ind DP are likewise expressible, so MIRAGE subsumes the heuristics.

6.2 Scaling to a City via Density-Adaptive Local LPs

Program (4) has $|V|^2$ variables and $|V|^2$ adversary constraints (4b)—about 10^7 and 10^7 for our graphs—and is intractable directly, which is exactly why prior work stayed on small grids. MIRAGE decomposes the graph into density-adaptive local regions and solves one small LP per region (Algorithm 1). Starting from the highest-prior unassigned node, each region greedily takes its C nearest still-unassigned nodes, so dense areas are seeded first and covered by tighter regions. Within a region of size C the LP has $O(C^2)$ variables and $O(C^2)$ constraints and solves in milliseconds; there are $\lceil |V|/C \rceil$ regions, for an $O(|V|C^3)$ offline precomputation (measured in Section 10), after which release is an $O(1)$ categorical draw.

Why the decomposition is near-lossless. Releasing far from the true node is never optimal under a finite distortion budget: it spends utility without buying proportional privacy, since the adversary's error saturates at the prior's spread. The optimal global mechanism therefore keeps $f(o \mid v)$ supported near v; partitioning V into local regions of radius comparable to the budget preserves that support. Empirically (Section 10) enlarging C raises privacy only marginally beyond the knee, confirming little is lost. The partition's one side effect—revealing region membership—matters only for the trajectory adversary and is analysed honestly in Section 13.

6.3 Trajectory-Aware Variant (MIRAGE-T)

The snapshot LP optimises against the static prior π. Against the trajectory adversary, however, the effective prior at step t is the mobility-propagated belief b_t^- of (3), which is far more concentrated. MIRAGE-T solves, at each step, the region LP with prior $b_t^-|_R$ instead of π_R (receding-horizon / certainty-equivalent optimal control): it hides where the adversary is currently confident. We evaluate whether this closes the gap in Section 13.

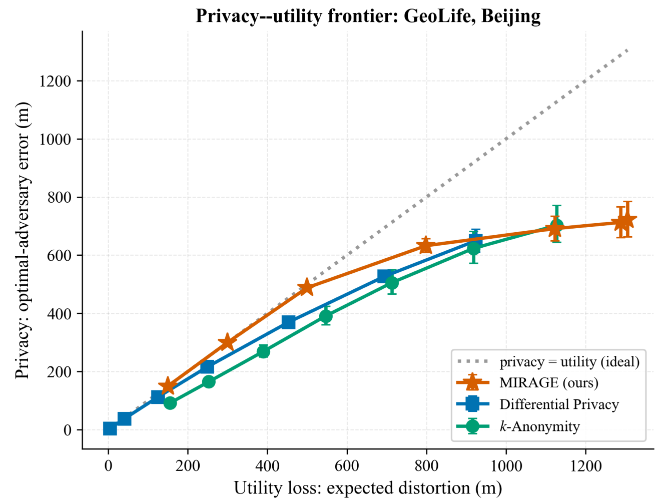

Fig. 2. Privacy–utility frontier (GeoLife, real graph, 95% CI). MIRAGE (optimal) dominates the heuristics; the vertical gap to MIRAGE is the price of heuristics. All converge at the intrinsic uncertainty ceiling.

7 Experimental Setup

All mechanisms share the graph, data, prior, and distance layer, and we evaluate in two complementary ways. (i) The frontier: for each mechanism we compute privacy (optimal-adversary error) and utility (expected distortion) analytically on the same regions, sweeping the mechanism's knob ($D_{\max}$ for MIRAGE, ε for DP, k for k-anonymity), with 95% weighted-bootstrap confidence intervals over regions. This isolates mechanism quality from sampling noise and matches the LP objective. (ii) A deployment comparison: all seven mechanisms (the five classical ones, the density-adaptive heuristic DA-Hybrid, and MIRAGE) at representative operating points, under the sampled snapshot and trajectory adversaries, reporting availability and spatial error. The energy model is swept across radio technologies (Section 14). Unless noted MIRAGE uses C=28. LPs are solved with the HiGHS solver; the shared distance matrix (one dense $|V| \times |V|$ array) lets the exact Bayesian adversary run at $\sim$3 k nodes.

8 Results I: The Price of Heuristics

Fig. 2 shows the privacy–utility frontier on the real GeoLife graph. MIRAGE hugs the ideal line (privacy = utility) at low distortion and dominates DP and k-anonymity throughout the practical regime; all mechanisms converge at high distortion to the network's intrinsic prior-uncertainty ceiling ($\sim$720 m, the maximum error any mechanism can induce given the prior). Table 4 quantifies the gap: at matched utility MIRAGE delivers +13 to +22% more privacy than the best heuristic in the 150–500 m regime on GeoLife—that fraction of their distortion is wasted by DP and k-anonymity. The gap closes only past $\sim$1100 m, where all saturate. A region-level paired bootstrap (113 regions) confirms the advantage is statistically significant, not confidence-interval overlap: at matched utility ($\sim$500 m) MIRAGE is more private than DP by +118 m (95% CI $[+109, +128]$, one-sided $p < 0.001$) and than k-anonymity by +97 m ($[+65, +126]$, $p < 0.001$).

Cross-city generalisation. The result holds on all three datasets and both cities (Fig. 3): the practical-regime gain is

TABLE 4
Price of heuristics (GeoLife, real graph): privacy (optimal-adversary error, m) at matched utility loss.

Utility (m)	MIRAGE	DP	k-anon	MIRAGE gain
150	150	133	92	+13%
300	300	254	201	+18%
500	488	400	355	+22%
800	632	583	555	+9%
≥1100	700+	649	700+	∼0% (saturated)

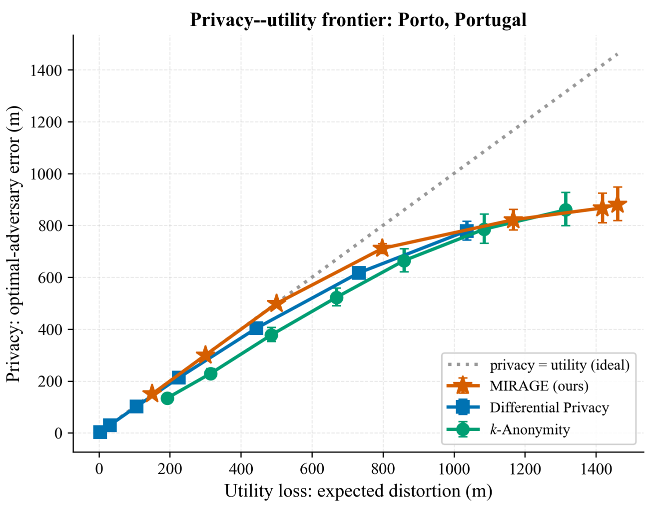

Fig. 3. Cross-city check on the real Porto (Portugal) network under the taxi dataset. MIRAGE dominates here too (+5–12%).

TABLE 5
Prior heterogeneity explains the cross-city gap. Lower within-region entropy (more skewed occupancy) ⇒ larger MIRAGE gain.

City	Prior ent.	Within-reg. ent.	Mob. cond. ent.	MIRAGE gain
GeoLife	0.80	0.79	0.30	20%
T-Drive	0.89	0.84	0.75	15%
Porto	0.74	0.87	0.19	10%

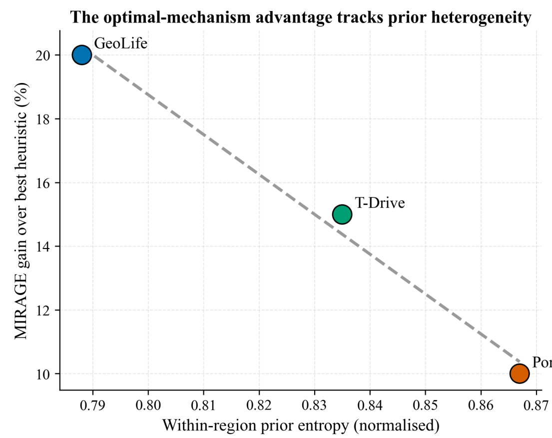

Fig. 4. The optimal-mechanism advantage tracks within-region prior heterogeneity across cities: more concentrated occupancy, larger gain.

+13–22% on GeoLife, +9–17% on T-Drive, and +5–12% on Porto. MIRAGE is Pareto-optimal and dominant everywhere; only the magnitude varies, which we now explain.

9 Results II: Why the Gap Varies

MIRAGE exploits the prior; DP and k-anonymity essentially ignore it (they hide near-uniformly over neighbours). Its advantage should therefore track how non-uniform occupancy is within a local region: where the prior is skewed there is structure to exploit, where it is flat the heuristics are already near optimal. Table 5 confirms this: the within-region prior entropy (normalised) orders the three cities monotonically with the measured gain—GeoLife is the most concentrated (0.79, gain 20%), Porto the most uniform (0.87, gain 10%), T-Drive between (0.84, 15%)—while mobility predictability does not track the gap (as expected, since the snapshot frontier depends on the prior, not on transitions). Fig. 4 plots the relationship. This gives operators a rule of thumb: the more spatially concentrated their population, the more an optimal mechanism buys them.

10 Results III: Ablation and Scalability

Table 6 ablates MIRAGE's two design choices. Region size C trades privacy for compute: privacy rises with C (at $D_{\max}$=800: 421 m at C=12 to 715 m at C=40) while the whole-graph precompute grows as $O(|V|C^3)$ (1.3 s to 20.8 s); the knee is at C=28, and the marginal privacy above it confirms the local decomposition loses little. The geo-indistinguishability constraint (5) costs some optimality (632 → 456 m at ε/m=0.01) in exchange for a formal DP guarantee—quantifying the price of that guarantee within one framework. All precomputes complete in seconds to tens of seconds on a laptop for a ∼3 k node city, and release is $O(1)$, so MIRAGE is deployable.

How much does the decomposition cost? We measure it exactly on a small graph where the global $|V|^2$ LP is solvable: we compute the true global optimum (one LP over all nodes) and the local decomposition, both scored by the exact optimal-adversary metric (Fig. 5). In the operationally relevant regime the decomposition is essentially lossless—mean privacy gap 1.5% for distortion $\leq$ 500 m, 0% up to 300 m—so MIRAGE's measured dominance is genuinely near-globally-optimal. A gap appears only at high distortion (28–44% at $D_{\max} \geq 800$), where the global optimum releases across regions but the partition confines releases; but that is the saturated regime where all mechanisms converge to the uncertainty ceiling and the advantage has already vanished.

11 Results IV: Deployment Comparison

Table 7 compares all seven mechanisms on the real GeoLife graph under the sampled snapshot and trajectory adversaries. MIRAGE serves every user (100% availability), is on-graph, and provides strong balanced privacy; the two adversaries again reorder the heuristics. Repeating on T-Drive and Porto yields two robustness findings. First, rankings under the adversary metric hold across topology and city. Second, availability is dataset-density-driven, not mechanism-intrinsic: on dense T-Drive every mechanism reaches 100% availability, whereas on sparse GeoLife k-anonymity (33%) and density-aware (61%) suffer denial of service (Fig. 6). A mechanism's

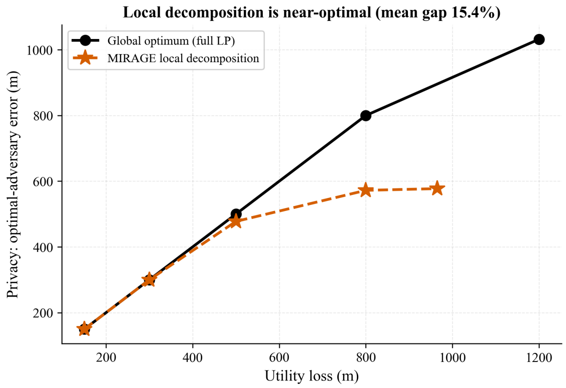

Fig. 5. The scalable local decomposition matches the global LP optimum in the practical regime (mean gap 1.5% for distortion ≤ 500 m), diverging only at high distortion where all mechanisms saturate.

TABLE 6
MIRAGE ablation (GeoLife, real graph). Privacy at $D_{\max}$=800 m and whole-graph precompute time vs. region size C.

Region size C	12	20	28	40
Regions	263	158	113	79
Privacy (m)	421	551	632	715
Precompute (s)	1.3	3.1	7.0	20.8

availability, then, is a property of the deployment as much as of the algorithm.

MIRAGE's own deployment point is consistent across all three settings (Table 8): 100% availability, on-graph output, and balanced snapshot/trajectory privacy, with error tracking each network's granularity.

12 Results V: Robustness to a Misspecified Prior

MIRAGE exploits the prior π, which a deployer must estimate from finite, possibly stale data. Does its advantage survive an imperfect estimate? We solve MIRAGE with a corrupted prior $\hat{\pi} = (1-\alpha)\pi + \alpha\mathbf{u}$ ($\mathbf{u}$ uniform, α the degree of ignorance: 0 perfect, 1 no prior at all) while the adversary attacks with the true π (Fig. 7). MIRAGE remains the most private for $\alpha \lesssim 0.25$; a deployer estimating occupancy from the same data operates far below this crossover. Under severe misspecification ($\alpha \geq 0.5$), however, MIRAGE can fall below the prior-free heuristics—it optimises for the wrong distribution while the adversary uses the right one. This is an honest deployment caveat and a precise one: the optimal mechanism is only as good as its prior, motivating a distributionally-robust variant (optimising the worst case over a prior-uncertainty set) as future work.

13 Results VI: Trajectory Adversary — an Honest Limit

MIRAGE is optimal against the snapshot adversary. Does that carry to the trajectory adversary? We stress-test it, and the answer—honestly—is no, for an instructive structural reason. Using the filtering adversary (3) on real user traces (Fig. 8), DP is the most trajectory-private mechanism at matched distortion: at 800 m distortion the tracking error is 676 m for DP versus ∼578 m for both MIRAGE variants. The cause is MIRAGE's density-adaptive partition—the very device that makes it scalable and snapshot-optimal. Because each release lies in a fixed region, observing it reveals region membership, which the sequential adversary exploits to localise; DP's diffuse, unpartitioned noise leaks less across time.

Our trajectory-aware variant MIRAGE-T (per-step LP against the mobility-propagated belief) helps, and instructively its benefit tracks mobility. On stationary GeoLife—where users linger, so the belief barely moves and the dynamic prior ≈ the static one—MIRAGE-T essentially matches snapshot-MIRAGE (∼578 vs. 577 m at 800 m distortion). On the mobile T-Drive taxis (low self-transition), where the belief shifts each step and there is real structure to exploit, MIRAGE-T improves trajectory privacy by +9 to +13% over snapshot-MIRAGE at matched utility and nearly closes the gap to DP (624 vs. 647 m at 800 m). So belief-aware release meaningfully helps exactly where trajectory tracking is most dangerous, but does not fully overcome the region-membership leak. This is a clean instance of the paper's thesis—privacy is threat-model dependent, here even for an optimal mechanism—and it isolates the key open problem: a scalable graph mechanism jointly optimal against snapshot and trajectory adversaries, plausibly via soft or overlapping regions whose releases do not reveal membership.

Can we un-leak the membership? A structural tension. We tested the natural fix directly: replace the single hard partition with a random mixture of two overlapping partitions, so a release can originate from either partition's region and no longer reveals one region (Fig. 9). It helps only marginally—by +2 to +7% on the mobile T-Drive traces and slightly hurts on the stationary GeoLife ones—and in neither case does it close the gap to DP (at ∼800 m distortion, 527–564 m for MIRAGE variants vs. 669 m for DP). The lesson is that the tension is structural: light de-partitioning is insufficient, because the same local region structure that yields snapshot-optimality inherently concentrates releases; matching DP's trajectory-robustness appears to require genuinely diffuse support, which

TABLE 7
Deployment comparison (GeoLife, real graph, 10 min). Privacy = Bayesian- adversary error (m, higher = more private).

Mechanism	Snap. AE	Traj. AE	Avail.	Loc. Err
k-Anonymity	1694	3459	33%	4493
Differential Privacy	730	947	100%	466
Graph-Constr. DP	1339	732	100%	1400
Density-Aware	1437	3459	61%	2379
DA-Hybrid (heuristic)	311	1885	100%	563
Temporal Cloaking	2577	4636	100%	3215
MIRAGE (ours)	846	855	100%	1090

TABLE 8
MIRAGE deployment point across cities/datasets (real graphs, 10 min, $D_{\max}$=800). Privacy = Bayesian-adversary error (m).

Dataset	Snap. AE	Traj. AE	Avail.	Loc. Err
GeoLife (Beijing)	846	855	100%	1090
T-Drive (Beijing)	645	481	100%	798
Porto (Portugal)	699	1904	100%	798

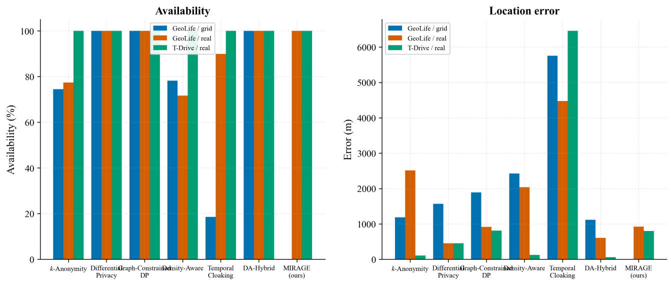

Fig. 6. Availability and error across the grid, the real graph (GeoLife), and the real graph under T-Drive. Availability rises to 100% for all mechanisms on the dense dataset—it is density-driven.

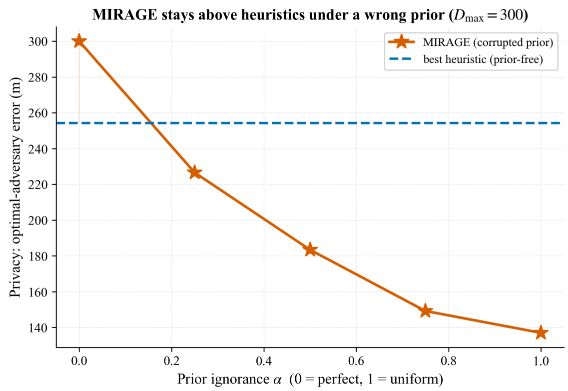

Fig. 7. Robustness to prior misspecification. MIRAGE stays above the best prior-free heuristic while the prior is reasonably accurate ($\alpha \lesssim 0.25$); a badly wrong prior can make it underperform.

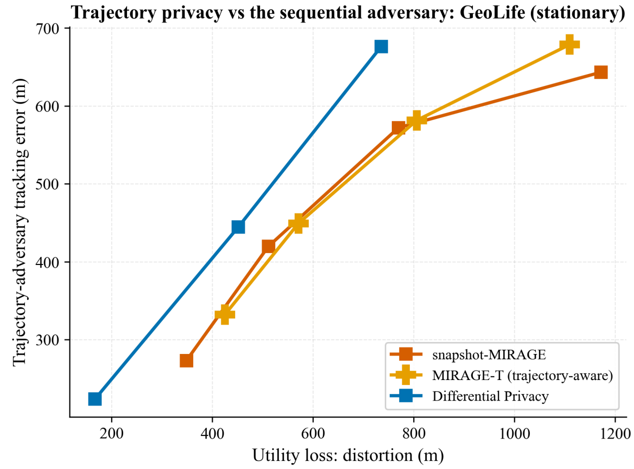

Fig. 8. Trajectory adversary (GeoLife, real traces). At matched distortion DP is most trajectory-private; MIRAGE's region partition leaks membership, and the mobility-aware MIRAGE-T does not close the gap.

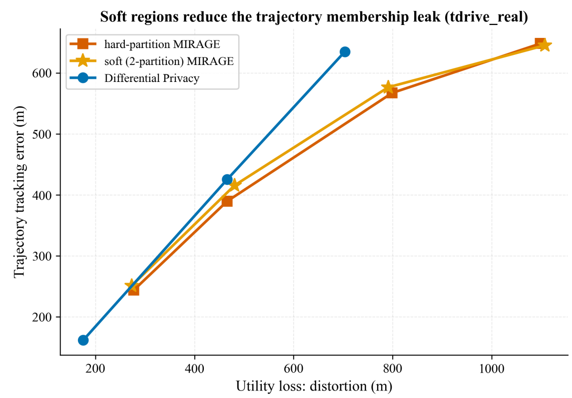

Fig. 9. Attempting to close the trajectory gap with soft (two-partition) regions. It helps marginally on mobile data and not at all on stationary data; DP remains more trajectory-robust—the membership-leak tension is structural.

sacrifices the snapshot optimality. A mechanism that is jointly optimal for both threats—perhaps a many-partition ensemble or a continuous-support graph LP—remains the central open problem this analysis sharpens.

14 Results VII: Energy-Model Sensitivity

Per-report energy is $E = E_{\text{radio}} + E_{\text{comp}} + E_{\text{retrans}}$, with $E_{\text{radio}} = P_{\text{tx}} t_{\text{tx}}$. Rather than fix one radio, we sweep BLE, Zigbee, LoRa, NB-IoT, and LTE-M (Fig. 10). The common claim that radio transmission dominates holds only for LPWAN/cellular radios (99% radio share for LoRa/NB-IoT/LTE-M) but not for BLE, where computation is 6–50% of energy for BFS-based mechanisms. MIRAGE's release-time cost is a single categorical draw ($O(1)$), so it is as radio-dominated as DP; its LP cost is paid once, offline.

15 Discussion

Deployment guidance. For a chosen utility budget on the operator's own road network and mobility statistics, MIRAGE gives the release policy that provably maximises snapshot-attacker error—up to +22% more privacy than tuned heuristics at equal utility, with 100% availability and on-graph outputs, and the gain is largest where the population is spatially concentrated (Section 9). Where a formal DP guarantee is required, the geo-indistinguishability-constrained variant applies at a measured cost. Where sequential tracking is the dominant threat, a diffuse mechanism (or the open membership-hiding design of Section 13) is preferable—the choice must be made against the relevant adversary, which our two-threat metric makes explicit.

Threats to validity. (i) Absolute error is graph-granularity dependent, so cross-topology conclusions rest on adversary-privacy and availability, not raw error. (ii) The energy model is analytical, suited to ranking rather than absolute prediction; we report its radio sensitivity explicitly. (iii) The mobility prior is first-order Markov; a richer prior would strengthen the trajectory adversary and can only lower reported trajectory privacy—so our trajectory numbers are, if anything, optimistic for all mechanisms. (iv) The local decomposition caps MIRAGE's reach at high distortion, where heuristics catch up at the uncertainty ceiling; hierarchical regions would extend the advantage. (v) MIRAGE's optimality is prior-conditioned (Section 12): a badly misspecified prior can erase its advantage, so a deployer must estimate occupancy reasonably well or use a distributionally-robust variant.

Ethical use. The framework is defensive: it helps operators release less information for a given utility, and it exposes when a mechanism is weaker than its nominal parameter

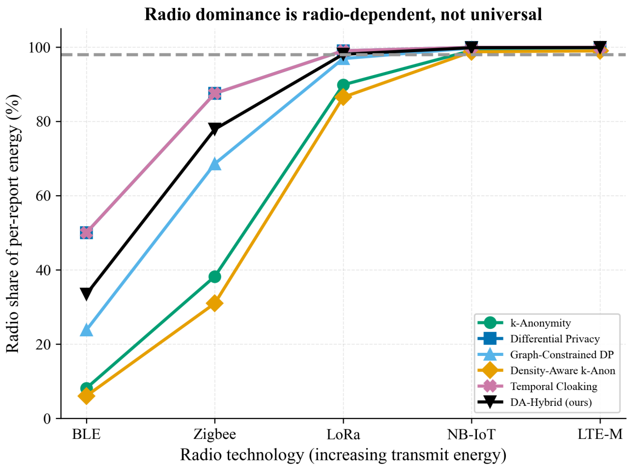

Fig. 10. Radio share of per-report energy vs. radio technology. Radio dominance is radio-dependent, not universal.

suggests. The adversary models are standard and use only public background knowledge.

16 Conclusion and Future Work

We measured location privacy on a common, adversary-grounded, threat-model-aware scale, and brought optimal obfuscation to a real road network at city scale with MIRAGE. On real networks of two cities across three datasets, deployed heuristics leave 5–22% of achievable privacy on the table in the practical regime, while MIRAGE is Pareto-optimal with full availability and on-graph outputs; the advantage is explained by prior heterogeneity, rankings hold from grid to real topology, and availability is shown to be dataset-density-driven. We also showed, honestly, that snapshot-optimality does not imply trajectory-optimality: the scalability partition leaks region membership. The central open problem—a scalable graph mechanism jointly optimal against snapshot and trajectory adversaries via membership-hiding soft/overlapping regions—together with richer mobility priors and hardware-measured energy, defines our future work. The complete framework is released for reproducible extension.

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